



AI Product/Engineer Intern Assignment

Project Title

AI-Powered Sports Engagement Content Agent

Objective

Develop an AI agent capable of generating engaging, multi-format sports content for Instagram quizzes, polls and more. The system should leverage web search to ensure factual accuracy and fresh content, and should support multiple engagement formats beyond simple MCQs.

Problem Statement

Traditional sports content on social media is largely limited to news, highlights, and opinion-based posts. Even interactive formats are often limited to a single question type (MCQ), which becomes repetitive for followers over time.

The goal of this project is to build an agent that generates **multiple types of interactive content** quizzes, true/false challenges, opinion polls, and fill-in-the-blank prompts so content creators can vary their engagement strategy, keep followers interested, and match the right format to the right platform surface (Story, Feed, Reel caption).

All content generated is **text-based** no image generation or image sourcing is required. Output should be ready to drop into Instagram's native text/sticker tools (Quiz sticker, Poll sticker, caption text, etc.).

Functional Requirements

- Select a **sport** (e.g., Cricket, Football, Tennis, Badminton, Basketball, etc.)
- Choose a **difficulty level** (Easy, Medium, Hard)
- Choose a **question/content type**:
 - Multiple Choice Question (MCQ)
 - True / False
 - This-or-That Poll (no single correct answer — pure opinion/engagement)
 - Fill in the Blank (with 4 answer options)
 - Guess the Number
- Generate a **batch** of 4–5 items per request, optionally mixing types in one batch
- Regenerate any individual item or the full batch on request
- Produce factually accurate content using retrieved knowledge, except for This-or-That polls, which are opinion-based by design



AI Agent Requirements

- Use web search to gather recent or fast-changing sports information (recent match results, records, transfers, tournament outcomes)
- Combine retrieved context with the LLM to generate high-quality content for each format
- Avoid hallucinations by grounding all factual claims in retrieved information; the agent should be able to cite which source (web search or vector DB) supported each answer
- Generate fresh and diverse content on every request — avoid repeating the same question/fact across sessions
- Apply a type-specific generation template for each content type (an MCQ prompt template, a True/False template, a Poll template, etc.) rather than one generic prompt for everything
- Validate every generated item against a schema for its type before returning it (e.g., MCQ must have exactly 4 options and 1 correct answer; Poll must have exactly 2 options and no correct answer; Guess-the-Number must have a numeric target plus an acceptable range)

Expected Output

1. Multiple Choice Question (MCQ)

- Sport Name
- Difficulty Level
- Question
- Four answer options (A, B, C, D)
- Correct answer
- Short explanation

2. True / False

- Sport Name
- Difficulty Level
- Statement
- Correct answer (True/False)
- Short explanation

3. This-or-That Poll

- Sport Name
- Prompt (e.g., "Messi or Ronaldo — who's the greater dribbler?")
- Two options
- No correct answer (flagged as opinion-based, not fact-checked)

4. Fill in the Blank

- Sport Name
- Difficulty Level
- Sentence with a blank
- Four answer options
- Correct answer



- Short explanation

5. Guess the Number

- Sport Name
- Difficulty Level
- Question (e.g., "How many runs did Virat Kohli score in the 2023 World Cup?")
- Target number
- Accepted tolerance range (e.g., ± 5)
- Short explanation

Expected Deliverables

- Working AI-powered content generation dashboard supporting all five content types
- Integration with web search for fresh/recent facts
- ChromaDB-based knowledge retrieval for stable/historical facts
- Type-specific prompt templates and output schema validation
- Batch generation with mixed content types and per-item regeneration
- Clean and well-documented code
- README with setup instructions, project overview, and a description of the type-specific architecture

Evaluation Criteria

- Accuracy of generated factual content (MCQ, T/F, Fill-in-blank, Guess-the-Number)
- Quality and relevance of retrieved information (web search + ChromaDB)
- Correct schema/format handling across all five content types
- Effective use of the AI agent (grounding, low hallucination rate, source citation)
- Diversity and freshness of content across repeated requests
- User experience and usability of the dashboard
- Code quality and documentation

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